

General Exam Summer 2007

Algebra

August 20, 2007

Rules

1. This is a closed book exam. To use a result, you can cite it by name (e.g., say "by the Main Theorem on Finitely Generated Modules over PIDs") or, if the result does not have a common name, you can just restate it (e.g., say "We know from class that the polynomial ring over a UFD is a UFD").
2. Make sure that I can understand what you write. It will help if you use complete sentences to communicate your ideas. I do not engage in reading minds. You really have to tell me what is going on. Make sure that it is impossible to misunderstand your write-up. I can be somewhat dense, and that will work to your disadvantage.
3. If you think a problem needs clarification, please ask. I will respond to the class.
4. There is a total of 50 points on this exam.
5. This exam has 9 problems and 10 pages. Make sure that none are missing in your copy.

Hints

1. Problems are not sorted according to difficulty.

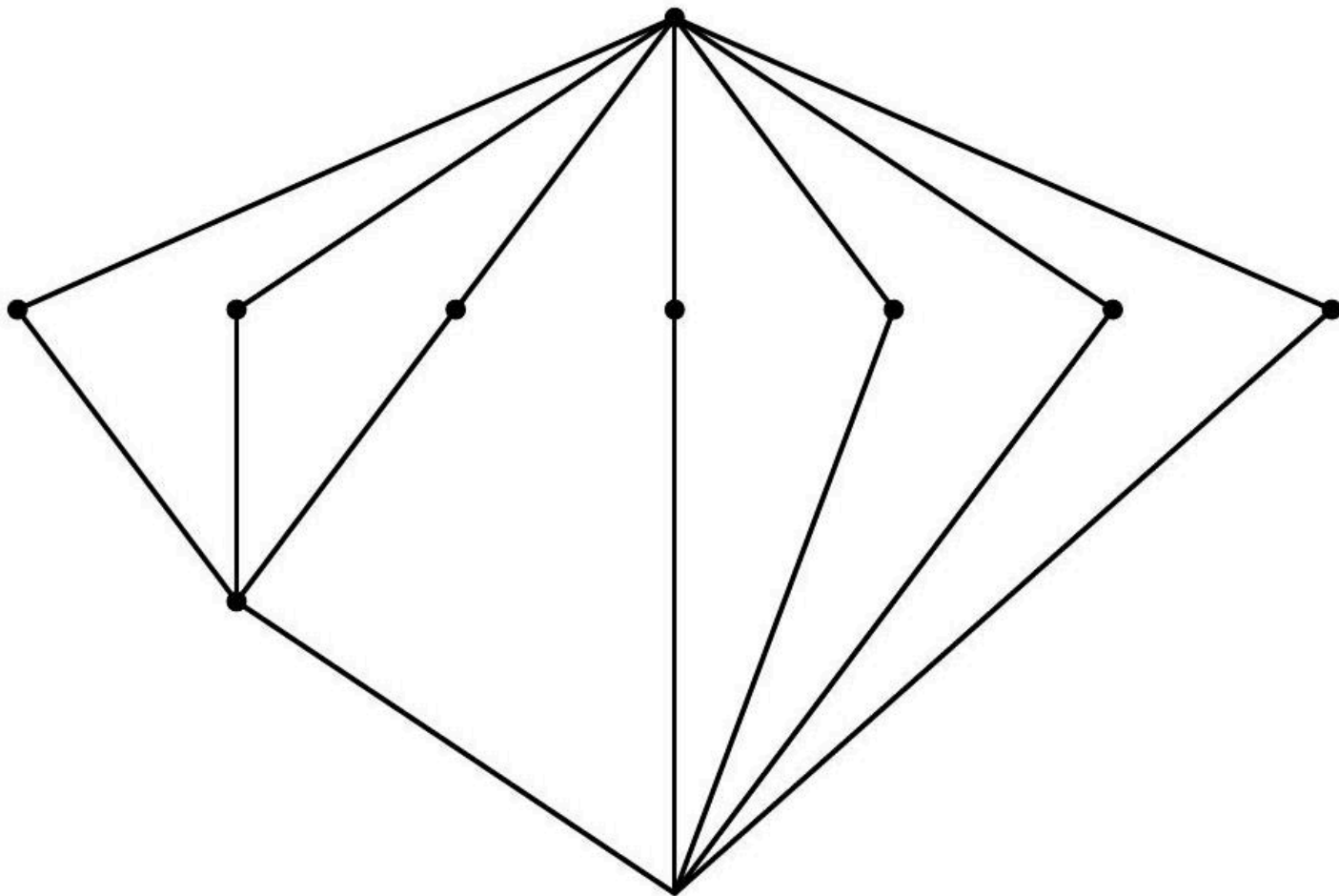
Room for your pledge:

Problem 1 [5 points]

Prove or disprove: the alternating group A_{2007} has a generating set that consists entirely of elements of order 2007.

Problem 2 [5 points]

Let M/K be a Galois extension of degree 12. The following is the lattice of intermediate extensions.



The lattice has the top field M , the bottom field K , seven vertices on an upper intermediate level, and one vertex on a lower intermediate level. Name the seven upper vertices U_1 through U_7 from left to right, and the lower intermediate vertex V . Each U_i is joined upward to M . The vertices U_1 , U_2 , and U_3 are joined downward to V , and V is joined downward to K . Each of U_4 , U_5 , U_6 , and U_7 is joined directly downward to K . These are all the edges shown. The auxiliary names U_i and V describe positions in the picture; the original vertices are unlabeled.

Here, vertices of the same height may correspond to intermediate extension of different degree. The top vertex corresponds to M and the bottom vertex corresponds to K . Show that the Galois group of M/K is A_4 .

Problem 3 [5 points]

The aim of this problem is to show that finite non-abelian simple groups do not arise as subgroups of $GL_2(\mathbb{C})$. You may use the Feit-Thompson theorem that says that finite non-abelian simple groups have even order.

1. Show that there is exactly one element of order 2 in $SL_2(\mathbb{C})$ and determine that matrix.

2. Show that any finite non-abelian simple subgroup of $\mathrm{GL}_2(\mathbb{C})$ is contained in $\mathrm{SL}_2(\mathbb{C})$.
3. Deduce that $\mathrm{GL}_2(\mathbb{C})$ does not contain any finite non-abelian simple subgroups.

Problem 4 [5 points]

Let $M := \mathbb{C}(t)$ be the rational function field, and consider the two $\mathbb{C}$ -automorphisms φ and ψ of M defined by $\varphi(t) = it$ and $\psi(t) = t^{-1}$.

1. Determine the group $G := \langle \varphi, \psi \rangle \leq \mathrm{Aut}_{\mathbb{C}}(M)$ generated by φ and ψ .
2. Let $K := \mathrm{Fix}_M(G)$ be the fixed field of G . Show that $G = \mathrm{Aut}_K(M)$ and that $t^4 + t^{-4}$ generates K as a field extension over $\mathbb{C}$.

Problem 5 [5 points]

Let A be an $n \times n$ square matrix with rational coefficients. Suppose that A has order 5 and assume that the equation $A\mathbf{v} = \mathbf{v}$ is only satisfied for $\mathbf{v} = 0$. Show that n is divisible by 4.

Problem 6 [5 points]

Show that there is some $n \in \mathbb{N}$ so that $\mathbb{Z}[in]$ is not a PID.

Problem 7 [5 points]

Show that

$$p(x) = x^3 - 3x^2 + 15x - 7$$

is irreducible in $\mathbb{Q}[x]$.

Problem 8 [5 points]

Let M be the $\mathbb{Z}$ -submodule of $\mathbb{Z}^2$ generated by

$$\begin{pmatrix} 2 \\ 10 \end{pmatrix}, \quad \begin{pmatrix} -2 \\ 8 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

Find a basis for M and determine the structure of $\mathbb{Z}^2 / M$ as an Abelian group.

Problem 9 [10 points]

For the following, no reasoning is required:

1. True or false: in a PID, irreducible elements are prime.
2. True or false: in a domain, irreducible elements are prime.
3. True or false: in a PID, prime elements are irreducible.
4. True or false: in a domain, prime elements are irreducible.
5. True or false: a solvable-by-solvable group is solvable.
6. Compute $(\mathbb{Z}_2 \oplus \mathbb{Q}) \otimes_{\mathbb{Z}} (\mathbb{Z}_2 \oplus \mathbb{Z}_5 \oplus \mathbb{Z})$.
7. State the definition of a prime ideal.
8. State Hilbert's basis theorem for polynomial rings.
9. State the definition of a solvable group.
10. State the definition of a Euclidean ring.